
Build the Geometry In, Don’t Hope It Emerges: A Reflection-Symmetry Audit of a Skeleton World Model

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Abstract

1 Bilateral (left–right) reflection is the most basic geometric symmetry of the artic-
2 ulated human body: mirroring a skeleton across the sagittal plane and swapping
3 left/right landmarks yields another valid skeleton, so the operation generates a
4 $\mathbb{Z}/2$ group acting on skeleton space. We ask whether a self-supervised skeleton
5 world model—a masked-latent-infilling S-JEPA over 33-joint pose sequences—
6 encodes this symmetry as a decodable, sign-flipping *laterality* axis. We turn the
7 question into an exact group-action equivariance test: a signed left-minus-right
8 motion target that is antisymmetric under the mirror by construction ($y(Mx) =$
9 $-y(x)$), probed under source-video-disjoint, repeated cross-validation against a
10 raw-coordinate ceiling, an untrained-encoder floor, a side-blind pooled control,
11 and a landmark-missingness control. On the frozen curriculum encoder the answer
12 is a **robust informative null**: the learned laterality feature reaches only
13 $R^2 = 0.198$ (stability interval $[0.175, 0.221]$)—below the untrained floor in
14 repeated-CV mean ($0.245 [0.214, 0.275]$; the stability intervals overlap, a partition-
15 stability heuristic rather than a significance test)—with a mirror slope of -0.70 in-
16 stead of the required -1 ; all three pre-registered gates fail. We then show the
17 geometry can be **recovered by construction**: a frame-averaged read-out over
18 $\{I, M\}$ is exactly antisymmetric (mirror slope -1.0000 for any nonzero weights)
19 and reaches $R^2 = 0.273 [0.253, 0.294]$, above the free read-out in repeated-CV
20 mean with disjoint stability intervals (again a stability heuristic, not a formal test);
21 a frame-averaged *encoder* wrapper is exactly token-equivariant with zero retrain-
22 ing (equivariance error 0.0). Finally, in a single-seed ablation, reflection *augmen-*
23 *tation* during pretraining does **not** induce the property (augmented axis 0.062 , not
24 clearing floor 0.078 ; slope no closer to -1). The lesson is a design principle
25 for geometry-aware world models of articulated bodies: *build the symmetry in; do*
26 *not hope it emerges*. All results are transductive (internal validity only); the source
27 video is the independent unit; dataset folder labels are annotations, not diagnoses.

1 Introduction

29 Physical-world scene understanding increasingly relies on *world models*—self-supervised predic-
30 tors that learn latent dynamics without labels. For articulated bodies, a natural question is whether
31 such models internalize the body’s **geometry**. The most elementary geometric fact about a human
32 skeleton is bilateral symmetry: reflecting it across the sagittal plane and relabeling left as right pro-
33 duces a physically valid skeleton. This is not a nuisance transformation to be averaged away; it is a
34 *structural prior* that a competent world model of an articulated body might be expected to respect.

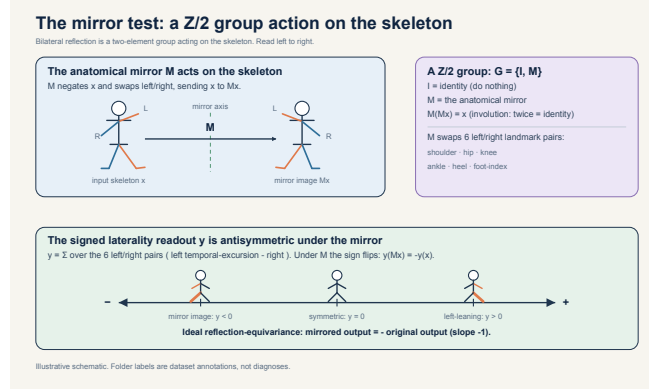


Figure 1: Bilateral reflection as a $\mathbb{Z}/2$ group action. The mirror M negates the sagittal coordinate and swaps left/right landmarks; the signed laterality target is antisymmetric by construction, $y(Mx) = -y(x)$, so “encoding the symmetry” means a decoded mirror slope of exactly -1 .

We study whether a frozen skeleton world model—an S-JEPA encoder trained by masked latent infilling over 33-joint BlazePose [Grishchenko et al., 2022, Google AI Edge] sequences—has learned bilateral symmetry in a form that is *decodable* and *sign-correct*. We make the question exact by treating the mirror as a group action. The mirror operator M (negate x ; swap all sixteen bilateral left/right landmark pairs—a valid whole-body reflection) satisfies $M^2 = I$, so $\{I, M\} \cong \mathbb{Z}/2$. We define a scalar **signed laterality** target measuring which side of the body moves more; by construction it is antisymmetric, $y(Mx) = -y(x)$. The world model “knows” bilateral symmetry iff its decoded laterality flips sign when the input is mirrored—a linear mirror slope of -1 —while remaining decodable at all (Fig. 1).

Contributions.

1. **A reusable geometry audit** for skeleton world models: an exact $\mathbb{Z}/2$ mirror test with a raw ceiling, an untrained floor, a side-blind control, and a missingness control, evaluated under source-disjoint *repeated* cross-validation with pre-registered gates (Fig. 2).
2. **An informative null with uncertainty.** The audited curriculum checkpoint *fails* the audit: the learned laterality feature does not beat the untrained floor under repeated CV, and the mirror slope is -0.70 , not -1 (§5, Fig. 3). We characterize the failure—an alpha-sensitivity sweep shows the feature is decodable only under heavy regularization, consistent with (but not proving) a weak, collinear geometry.
3. **A constructive fix by group averaging.** A frame-averaged read-out is antisymmetric by construction (slope -1.0000) and beats the free read-out in repeated-CV mean with disjoint stability intervals (a partition-stability heuristic); a frame-averaged encoder is exactly token-equivariant with zero retraining (§6–§7, Figs. 4–5).
4. **A negative result for augmentation.** In the audited single-seed Stage-0 arm, reflection augmentation during pretraining did not induce the symmetry (§7.2), sharpening the design principle: *build it in, do not hope it emerges* (Fig. 6).

We connect throughout to the workshop’s themes: this is a **geometry-aware** probe (and fix) of a world model for an **articulated, deformable** body, whose target is a **proprioceptive** quantity—which side moves more.

2 Related work

Equivariant and geometric deep learning. Group-equivariant networks build symmetry into the architecture, from group-convolutions and steerable CNNs [Cohen and Welling, 2016, 2017, Weiler and Cesa, 2019, Weiler et al., 2018, Worrall et al., 2017] to equivariant graph and attention networks [Satorras et al., 2021, Fuchs et al., 2020], unified under the geometric-deep-learning program [Bronstein et al., 2021]. Rather than constrain every layer, *frame averaging* attains exact equivariance by

69 symmetrizing over a group (or a frame of it) around an arbitrary backbone [Puny et al., 2022], and
70 *canonicalization* learns to align inputs to a canonical pose [Kaba et al., 2023, Jaderberg et al., 2015].
71 A complementary line asks whether symmetry can instead be *learned from data* via augmentation
72 [Benton et al., 2020]. Our read-out and encoder constructions are frame averaging over $\mathbb{Z}/2$; our
73 augmentation arm is a direct test of the learned-invariance hypothesis.

74 **Self-supervised skeleton models.** Skeleton action recognition is dominated by spatio-temporal
75 graph networks [Yan et al., 2018, Shi et al., 2019]; recent self-supervised approaches use masked
76 prediction and joint-embedding objectives [Abdelfattah and Alahi, 2024, Mao et al., 2023, Xu et al.,
77 2024], echoing image/video JEPA [Assran et al., 2023, Bardes et al., 2024] and VICReg-style regu-
78 larization [Bardes et al., 2022]. These works optimize for downstream accuracy; none audit whether
79 the learned latent respects the body’s reflection symmetry as an exact, sign-flipping axis.

80 **Gait symmetry.** Clinically, left–right symmetry indices are established descriptors of healthy and
81 pathological gait [Sadeghi et al., 2000, Patterson et al., 2010, Robinson et al., 1987, Błażkiewicz
82 et al., 2014]. We borrow the *concept* of a signed bilateral asymmetry but use it purely as a geometric
83 decodability target on public pose data—not as a clinical measurement.

84 3 Bilateral symmetry as a group action

85 **Group and operator.** Let $x \in \mathbb{R}^{T \times 33 \times 3}$ be a pose sequence. The mirror M negates the x -
86 coordinate of every joint and applies the left/right permutation that swaps all sixteen bilateral
87 BlazePose landmark pairs—a valid whole-body reflection. Then $M^2 = I$ and $G = \{I, M\} \cong \mathbb{Z}/2$.

88 **Target.** For each of the six bilateral pairs (11, 12), (23, 24), (25, 26), (27, 28), (29, 30), (31, 32)
89 (shoulders, hips, knees, ankles, heels, foot-indices), let ℓ_k (resp. r_k) be the per-joint temporal stan-
90 dard deviation of the left (resp. right) landmark summed over x, y, z . The signed laterality target is
91 $y(x) = \sum_k (\ell_k - r_k)$. Mirroring swaps $\ell_k \leftrightarrow r_k$, so

$$y(Mx) = -y(x).$$

92 This is a group-action equivariance condition $y(T_g x) = \rho(g) y(x)$ with the sign representation
93 $\rho(I) = +1$, $\rho(M) = -1$. A model “encodes bilateral symmetry” iff a read-out of its features
94 reproduces this antisymmetry: **mirror slope** -1 , while being decodable (R^2 well above floor).

95 **Backbone.** The world model is a frozen S-JEPA (config: 64 frames, 33 joints, segment length 4,
96 embed dim 96, 4-layer encoder, 2-layer predictor, 4 heads; masked latent infilling with a VICReg
97 term), trained by a curriculum beginning with the normal-annotated rows and adding each annotated
98 condition. We bind all results to a single checkpoint (fingerprint 7d13841a . . .). The primary cohort
99 is the **626** sequences / **93** source videos the encoder actually trained on (fully transductive); a **642/94**
100 superset is reported only for robustness.

101 4 The audit protocol

102 **Five lanes** (Fig. 2). **A — learned:** per-pair $[\ell - r, \ell + r]$ token statistics from the frozen encoder. **B**
103 **— raw ceiling:** target from coordinates ($R^2 \approx 1$), a decodability sanity check. **C — floor:** identical
104 architecture, random weights. **D — pooled:** whole-body mean/std, side-blind (must stay low). **E —**
105 **missingness:** per-joint left/right valid-fraction (must stay low; if it rivals A, the learned score may
106 reflect left/right visibility asymmetry rather than motion).

107 **Estimator.** Source-disjoint GroupKFold; inner ridge-penalty selection over $\alpha \in$
108 $\text{logspace}(-3, 3, 13)$; per-fold standardization; SVD solver (used to reduce sensitivity to the
109 numerical ill-conditioning seen in the original run). We repeat the whole grouped CV over 10
110 reshuffles of the *fixed* 626-sequence / 93-source cohort and report the mean with a t -based interval
111 ($\text{df} = 9$, $t^* = 2.262$). Because every reshuffle re-partitions the same fixed rows, this is an
112 **across-reshuffle stability interval:** it quantifies sensitivity to the CV partition, not sampling
113 variability over a population, and carries no out-of-sample or generalization guarantee [Bengio
114 and Grandvalet, 2004, Roberts et al., 2017]. Interval overlap or non-overlap is therefore a stability
115 heuristic, not a formal significance test.

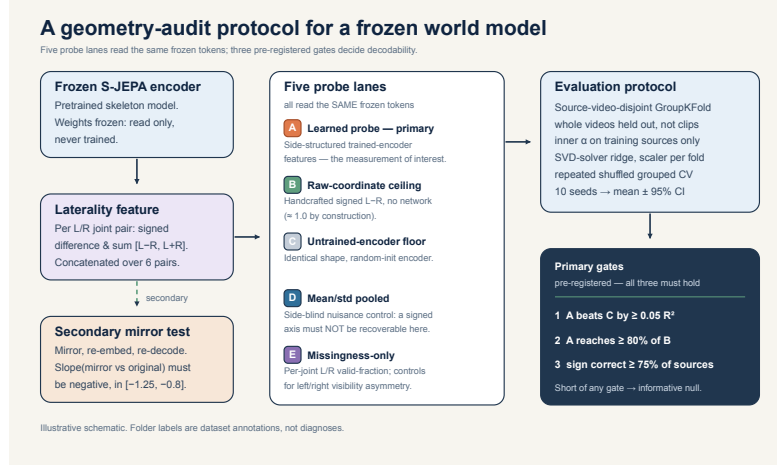


Figure 2: The five-lane audit. A learned per-pair token statistics; B raw-coordinate ceiling ($R^2 \approx 1$); C untrained-encoder floor; D side-blind pooled control; E landmark-missingness control. Estimation is source-video-disjoint GroupKFold repeated over ten reshuffles of a fixed cohort, with pre-registered gates.

Table 1: E1 laterality probe (626 primary, canonical checkpoint).

Lane	Single-partition R^2	Repeated-CV R^2 (stability interval)
A — learned	0.268	0.198 [0.175, 0.221]
C — untrained floor	0.229	0.245 [0.214, 0.275]
D — pooled (side-blind)	0.107	0.101 [0.089, 0.114]
E — missingness-only	0.162	0.202 [0.173, 0.231]
B — raw ceiling	≈ 1.000	—

Pre-registered gates. (i) A beats C by $\geq 0.05 R^2$; (ii) A reaches $\geq 80\%$ of B; (iii) sign correct on $\geq 75\%$ of held-out sources. **Secondary geometry band:** mirror slope negative and within [-1.25, -0.80].

5 E1 — the emergent null, hardened

Table 1 reports the five lanes on the primary cohort.

Mirror slope -0.703 (no sign flip); sign consistency $0.576 [0.548, 0.604]$. **All three gates fail.** The decisive observation is that the single-partition ordering $A(0.268) > C(0.229)$ *reverses* under repeated CV, $A(0.198) < C(0.245)$: the original notebook’s lone “win” was a single partition near the top of A’s spread (maximum across the ten CV reshuffles 0.272). An alpha sweep shows A’s score rising monotonically from -2.04 at $\alpha = 10^{-3}$ to $+0.268$ at $\alpha \geq 316$ —a feature decodable only under heavy regularization is consistent with a weak, collinear geometry, not a clean axis (Fig. 3). The 642 robustness cohort reproduces the pre-hardening numbers bit-for-bit ($A = 0.241$, $C = 0.190$, slope -0.627) and is the *only* configuration where gate (i) passes on a single partition; it does not survive cohort-matching plus repeated CV. The missingness control behaves consistently with the null: under repeated CV, Lane E ($0.202 [0.173, 0.231]$) has a repeated-CV mean close to learned Lane A (0.198), which flags possible visibility confounding rather than establishing equivalence—corroborating, not undermining, the null. (The single-partition `missingness_control_ok` flag, on which E sits below A, does not survive repeated CV, and E is itself estimator- and cohort-sensitive— 0.162 single vs 0.202 repeated on 626, and 0.047 on the 642 superset—so we read it only as corroboration.) **Verdict: a robust informative null**—the world model did not learn bilateral symmetry as a decodable, sign-flipping axis.

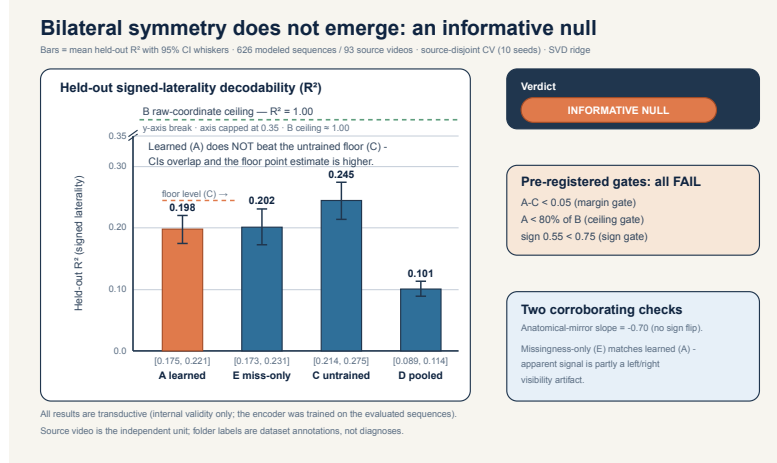


Figure 3: The hardened null. Under repeated CV the learned lane A sits below the untrained floor C in mean with overlapping stability intervals; the single-partition ordering $A > C$ reverses. The alpha sweep rises monotonically from -2.04 to $+0.268$, decodable only under heavy regularization—consistent with a weak, collinear feature.

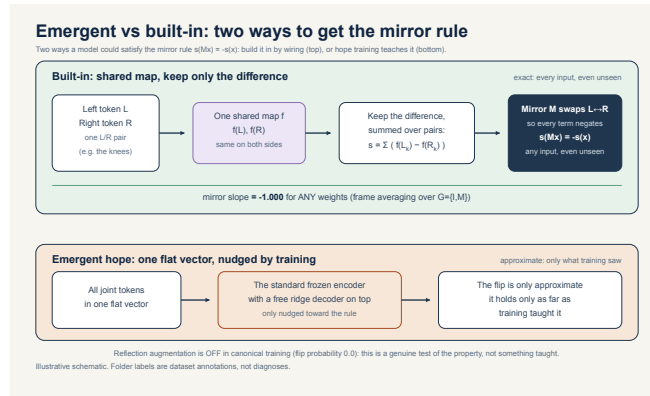


Figure 4: Frame averaging over $\{I, M\}$. The antisymmetric part $\Phi = A(x) - A(Mx)$ has decoded mirror slope exactly -1 for any nonzero weights; the symmetric part $\Psi = A(x) + A(Mx)$ has slope $+1$ and cannot predict an antisymmetric target.

137 6 E2 — reflection-equivariant read-out by construction

138 Rather than hope the frozen feature $A(x)$ is antisymmetric, we **average over the group** [Puny et al.,
 139 2022]. Define $\Phi(x) = A(x) - A(Mx)$ and $\Psi(x) = A(x) + A(Mx)$; then $\Phi(Mx) = -\Phi(x)$ and
 140 $\Psi(Mx) = \Psi(x)$. A *linear* read-out on Φ has mirror slope **exactly -1 for any nonzero weights**
 141 (Fig. 4).

142 Two observations stand out. **(1) Built-in beats free** in repeated-CV mean (0.273 vs 0.198; disjoint
 143 stability intervals—a partition-stability heuristic, not a formal test): imposing the geometry unlocks
 144 decodable structure the free read-out does not reach from the *same frozen encoder*— Φ additionally
 145 evaluates the mirrored input $A(Mx)$. **(2) The controls are consistent with the antisymmetric tie—not capacity or the extra pass—as the source of the gain, but they do not isolate a**
 146 **causal decomposition.** The symmetric Ψ cannot predict an antisymmetric target (0.015, slope $+1$;
 147 single-partition); a shared-map equivariant MLP is exactly antisymmetric and scores 0.243 (single-
 148 partition; linear Φ already optimal, nonlinearity adds nothing); an unconstrained $\geq$ -capacity MLP
 149 that sees both passes but is *untied* collapses to 0.047 (single-partition) with slope -0.430 . Decom-
 150 posing the point estimates, Φ exceeds free A by 0.075 in repeated-CV mean; the corresponding
 151 architecture and learned-encoder contrasts ($\Phi_{\text{floor}} - A$ and $\Phi - \Phi_{\text{floor}}$) are 0.021 and 0.054, with

Table 2: E2 read-out (626 primary). A, Φ (learned), and Φ (untrained) report repeated-CV means with stability intervals; Ψ is a single-partition diagnostic; mirror slopes are single-partition.

Read-out	R^2	Mirror slope
A — free ridge (repeated CV)	0.198 [0.175, 0.221]	−0.703
Φ — frame-averaged, learned (repeated CV)	0.273 [0.253, 0.294]	−1.0000001
Φ — frame-averaged, untrained (repeated CV)	0.219 [0.175, 0.264]	−0.99999996
Ψ — symmetric part (single-partition)	0.015	+1.000

Table 3: E3.2 augmentation ablation (270 normal-annotated rows, transductive). A_free reports repeated-CV means with stability intervals; free-readout slopes are single-partition.

Encoder	A_free R^2 (stability interval)	Free-readout slope
arm_on (flip 0.5)	+0.062 [+0.011, +0.114]	−0.510
arm_off (flip 0.0)	−0.049 [−0.113, +0.016]	−0.753
canonical Stage-0	−0.022 [−0.075, +0.031]	−0.553
untrained floor	+0.078 [+0.013, +0.144]	−0.818

overlapping intervals—so the *learning* benefit is suggestive, not decisive. Φ remains far below the raw ceiling (sign consistency 0.568); the constructed read-out is exactly antisymmetric but explains only part of the target variance on the audited cohort, motivating pushing symmetry into the encoder.

7 E3 — reflection-equivariant encoder

7.1 Frame-averaged encoder (zero-retrain, exact)

Averaging the *encoder* rather than the read-out, $E'(x)[t, j] = \frac{1}{2}(E(x)[t, j] + E(Mx)[t, \sigma(j)])$ with σ the left/right joint permutation, makes the frozen encoder **exactly token-level reflection-equivariant with no retraining**: token-equivariance error, diff-block antisymmetry error, and sum-block symmetry error are all 0.0 to machine precision. The laterality feature on E' splits exactly into an antisymmetric $\ell - r$ block (slope -1 for any nonzero read-out; $R^2 = 0.253$ [0.226, 0.280]) and a symmetric $\ell + r$ block. This antisymmetric block *corresponds to* E2’s Φ and realizes the same exact slope- -1 guarantee, but is not numerically identical: it equals (up to per-fold standardization) one half of Φ ’s $\ell - r$ sub-block, whereas Φ additionally antisymmetrizes the $\ell + r$ block—nonzero only because the original encoder is not itself equivariant—which is why Φ scores slightly higher (0.273 vs 0.253; overlapping stability intervals). Honest boundary: a *free* ridge on the full E' feature still gives slope -0.767 —encoder equivariance does not force an antisymmetric *decoder*.

7.2 Does reflection augmentation induce it? (a single-seed negative)

We retrain Stage-0 (the normal-annotated rows only, 270 sequences / 29 sources, 300 epochs) with one switch: sample-level consistent reflection augmentation on ($p = 0.5$) vs off (0.0), identical trainer, seed, and hardware [Benton et al., 2020]. Augmentation changes the objective (final JEPA loss 0.807 vs 0.720) but not the geometry (Table 3). The negative rests on the mirror *slope*, not on decodability: the augmented and untrained axes have similar repeated-CV means (0.062 [0.011, 0.114] vs 0.078 [0.013, 0.144]; overlapping stability intervals), and the free-readout slope does not move toward -1 . Strikingly, the **untrained** encoder has the slope *closest* to -1 (-0.818 , vs the augmented arm’s -0.510 , the *furthest*)—so in this single-seed arm slope proximity alone is not evidence of learned equivariance (it is consistent with sensitivity to feature geometry), which is precisely why the by-construction guarantee (exact -1 for any nonzero weights) is the right instrument. This arm is a single-seed proof-of-concept, so we report it as a single-seed negative rather than a population claim.

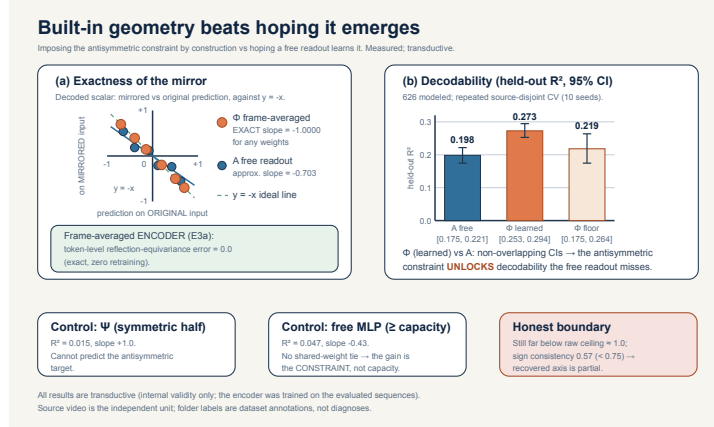


Figure 5: Built-in beats emergent. Left: the emergent read-out (E1) sits at or below floor with mirror slope -0.70 ; the frame-averaged read-out (E2) reaches 0.273 with slope exactly -1 . Right: the same contrast at the encoder—augmentation (E3.2) does not induce the axis, while the frame-averaged encoder (E3.1) is exactly equivariant.

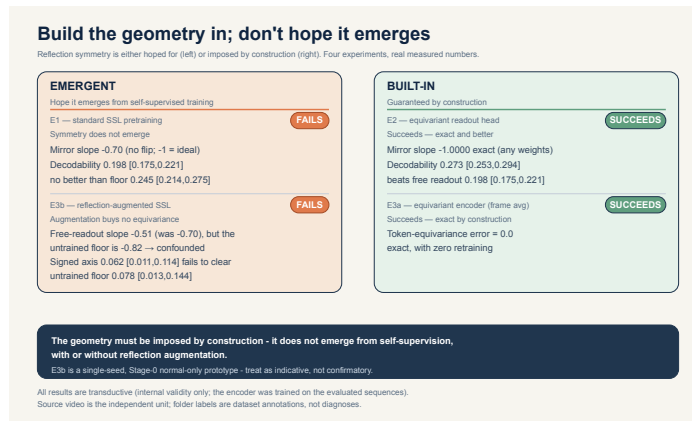


Figure 6: The design ladder for geometry-aware world models: hoping symmetry emerges (E1, E3.2) fails or is unreliable on the audited system; imposing it by frame averaging (E2, E3.1) makes it exact and cheap. *Build the geometry in; don't hope it emerges.*

182 8 Discussion: the emergent-vs-built-in ladder

183 The four experiments form a 2×2 grid (Fig. 6): symmetry in the *read-out* or the *encoder*, *hoped for*
 184 or *built in*. Emergent read-out (E1): slope -0.70 , learned $\approx$ floor, gates fail; built-in read-out (E2):
 185 slope -1.0000 , $0.273 > 0.198$ (disjoint intervals). Emergent encoder (E3.2): slope not toward -1 ,
 186 axis $\leq$ floor; built-in encoder (E3.1): token error 0.0 , exact block split.

187 Every row tells the same story (Fig. 5): in the audited system—this frozen curriculum checkpoint,
 188 and a single-seed reflection-augmentation arm—standard and reflection-augmented self-supervision
 189 leave bilateral symmetry un-learned, while frame averaging the read-out or encoder recovers it to
 190 machine precision with zero or minimal retraining. The constructive half of this finding is general
 191 (frame averaging imposes exact $\mathbb{Z}/2$ equivariance by construction); the negative half is scoped to
 192 what we audited. For geometry-aware world models of articulated bodies, the practical reading is
 193 that symmetry is cheap to *impose* and—at least here—not reliably obtained by *hoping* it emerges.

9 Limitations

Results are **transductive**—the encoder was trained on the evaluated sequences—so they carry internal validity only and do not speak to generalization to new sources; fold-local full retraining would be required for that. The signed target captures one scalar ($\mathbb{Z}/2$) symmetry, not the full kinematic tree; the augmentation arm is a single-seed prototype on the `normal`-annotated rows. The recovered axis, even built-in, remains well below the raw ceiling; the constructed read-out is exactly antisymmetric but explains only part of the target variance on the audited cohort. The curriculum’s later stages are label-informed, so “the world model” is not purely self-supervised across all stages.

10 Ethics and data-use

We study a public gait dataset (GAVD [Ranjan et al., 2025, GAVD project, 2026]) whose official distribution provides annotations and public YouTube video URLs rather than raw video files; users retrieve media independently and are responsible for YouTube’s terms of service, institutional ethics review, and applicable copyright, privacy, and data-protection rules. This analysis uses derived pose sequences and does not infer identity; no raw video or identity-bearing frames are redistributed. **The workspace does not currently contain a recorded institutional ethics determination or completed data-use review; both must be resolved before submission.** The dataset’s condition folders (`normal`, `parkinsons`, `stroke`, `myopathic`, `cerebralpalsy`) are **dataset annotations, not diagnoses**; the independent unit of analysis is the source video, not the individual, and no clinical claim is made or implied.

11 Conclusion

Bilateral symmetry is a natural geometry audit for skeleton world models. The audited S-JEPA fails it—for this checkpoint, bilateral symmetry does not emerge as a decodable, sign-flipping axis, and a single-seed reflection-augmentation arm does not induce it—but frame averaging recovers the axis by construction, exactly, at the read-out or the encoder. The constructive principle generalizes even where the negative is scoped to what we tested: for physical-world models of articulated bodies, *build the geometry in; don’t hope it emerges.*

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